

SUNWU, China

WMO: 505640

Lat: 49.4169N

Lon: 127.3314E

Elev: 235

StdP: 98.53

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.5	-32.1	-37.1	0.1	-34.2	-34.9	0.1	-31.9	7.3	-17.8	6.2	-16.4	0.7	220	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	30.7	20.4	28.9	20.3	27.3	19.6	23.0	27.1	22.0	26.3	21.1	25.3	4.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.9	25.1	20.7	15.8	24.3	19.7	14.8	23.3	69.9	26.9	66.1	26.3	62.7	25.4	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.6	7.4	6.3	-36.0	33.6	2.8	1.8	-38.0	35.0	-39.6	36.0	-41.1	37.1	-43.2	38.4		
			-36.1	24.6	2.8	1.1	-38.0	25.4	-39.7	26.0	-41.2	26.7	-43.2	27.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.1	-21.6	-17.3	-7.7	3.8	11.9	18.4	21.3	18.9	12.0	2.7	-10.2	-20.2
	DBStd	15.93	4.57	5.03	6.34	4.66	4.14	3.52	2.53	3.06	4.12	4.78	6.74	4.74
	HDD10.0	4308	979	763	548	193	26	0	0	0	25	231	607	937
	HDD18.3	6484	1237	997	806	436	203	42	4	29	192	485	857	1196
	CDD10.0	1062	0	0	0	7	85	252	351	277	87	4	0	0
	CDD18.3	196	0	0	0	0	5	43	96	48	3	0	0	0
	CDH23.3	1823	0	0	0	4	114	521	769	372	42	1	0	0
	CDH26.7	497	0	0	0	1	26	182	216	67	6	0	0	0
Wind	WSAvg	2.6	1.8	2.1	2.8	3.5	3.5	2.7	2.5	2.4	2.6	2.7	2.4	1.9
Precipitation	PrecAvg	547	4	4	6	24	47	87	146	123	70	21	10	6
	PrecMax	754	9	12	17	104	102	169	277	241	146	60	23	13
	PrecMin	359	0	0	0	2	17	22	25	28	14	0	0	1
	PrecStd	115	3	3	4	19	24	42	60	51	32	12	6	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.8	0.8	10.8	23.4	29.6	34.0	33.0	30.3	27.2	20.2	8.2	-4.8
		MCWB	-9.2	-2.2	3.6	10.9	15.1	19.4	21.3	22.0	18.0	11.5	3.5	-6.4
	2%	DB	-10.4	-2.6	7.3	18.8	26.7	31.0	30.7	28.4	24.4	16.0	4.5	-8.5
		MCWB	-11.8	-4.9	1.9	8.7	14.3	19.0	21.8	21.5	16.5	9.1	1.3	-9.8
	5%	DB	-12.5	-5.8	4.7	15.6	23.8	28.6	29.0	27.0	22.5	13.2	1.7	-10.8
		MCWB	-13.5	-7.7	0.2	7.3	12.9	18.4	21.2	20.6	15.3	7.5	-0.7	-11.9
	10%	DB	-14.3	-8.5	2.2	12.8	21.0	26.3	27.4	25.5	20.2	10.6	-0.6	-12.7
		MCWB	-15.1	-10.0	-1.2	5.6	11.9	17.6	20.8	19.7	14.2	5.7	-2.6	-13.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.0	-2.1	4.4	11.9	16.8	22.2	24.8	24.2	20.0	12.5	3.9	-6.2
		MCDB	-7.6	0.4	9.5	21.2	25.2	30.0	29.2	27.6	24.4	18.3	7.5	-4.9
	2%	WB	-11.6	-4.9	2.2	9.6	15.3	20.6	23.5	22.7	18.1	10.0	1.3	-9.7
		MCDB	-10.4	-2.6	6.8	17.3	23.5	27.4	27.7	26.5	22.0	14.5	4.2	-8.5
	5%	WB	-13.5	-7.7	0.5	7.8	14.0	19.6	22.5	21.7	16.7	8.1	-0.6	-11.8
		MCDB	-12.5	-5.8	4.1	14.8	21.8	25.8	26.6	25.5	20.8	12.3	1.5	-10.8
	10%	WB	-15.1	-10.0	-1.0	6.0	12.7	18.7	21.7	20.7	15.2	6.2	-2.6	-13.6
		MCDB	-14.3	-8.5	2.2	12.2	19.8	24.6	25.8	24.2	19.0	10.1	-0.7	-12.7
Mean Daily Temperature Range	5% DB	MDBR	12.0	14.1	12.6	12.0	12.8	11.8	10.0	10.5	12.7	11.1	10.3	10.7
		MCDBR	13.0	15.6	13.3	17.3	18.1	16.4	13.2	12.8	15.8	15.4	11.6	11.8
		MCWBR	11.8	13.3	8.7	9.4	8.2	6.5	5.5	6.3	8.1	9.3	8.6	10.6
	5% WB	MCDBR	12.8	15.4	12.3	15.8	15.4	12.9	10.3	10.3	12.7	13.5	11.2	11.6
		MCWBR	11.7	13.2	8.3	9.0	7.5	5.8	5.2	5.6	7.2	8.9	8.6	10.5
Clear-Sky Solar Irradiance	taub	0.281	0.309	0.375	0.491	0.558	0.519	0.495	0.442	0.403	0.396	0.326	0.288	
	taud	2.159	2.087	1.973	1.832	1.749	1.950	2.070	2.200	2.221	2.087	2.154	2.128	
	Ebn at Noon	750	815	812	751	717	755	768	793	784	697	676	672	
	Edh at Noon	79	109	147	191	216	178	156	131	115	110	80	71	
All-Sky Solar Radiation	RadAvg	1.77	2.99	4.44	4.93	5.41	5.71	5.32	4.79	3.92	2.62	1.70	1.35	
	RadStd	0.06	0.12	0.28	0.53	0.47	0.74	0.47	0.38	0.32	0.16	0.12	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page